

● HARD

● ALGEBRA

● ANSWER KEY

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# Linear functions

# 03

6 answers · Rationale-rich · Teach from doc

**Q1****D**

D.  $S = \frac{w}{85}$

Choice D is correct. It's given that  $w$  represents the total fence area, in square feet. Since the fence will be stained twice, the amount of stain, in gallons, will need to cover  $2w$  square feet. It's also given that one gallon of stain will cover 170 square feet. Dividing the total area, in square feet, of the surface to be stained by the number of square feet covered by one gallon of stain gives the number of gallons of stain that will be needed. Dividing  $2w$  by 170 yields  $\frac{2w}{170}$ , or  $\frac{w}{85}$ . Therefore, the equation that represents the total amount of stain  $S$ , in gallons, needed to stain the fence of the yard twice is  $S = \frac{w}{85}$ .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

**Q2****D**

D. The increase in the length, in centimeters, of the spring for each one-newton increase in the weight of the object

Choice D is correct. The value 2 is multiplied by  $w$ , the weight of the object. When the weight is 0, the length is  $30 + 2(0) = 30$  centimeters. If the weight increases by  $w$  newtons, the length increases by  $2w$  centimeters, or 2 centimeters for each one-newton increase in weight.

Choice A is incorrect because this describes the value 30. Choice B is incorrect because 30 represents the length of the spring before it has been stretched. Choice C is incorrect because this describes the value  $w$ .

### Q3

A

A. When  $f(x) = 0$ , the number is  $-\frac{19}{4}$ .

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Choice A is correct. It's given that the function  $f(x)$  is defined as 19 more than 4 times a number  $x$ . This can be represented by the equation  $f(x) = 4x + 19$ . The  $x$ -intercept of the graph of  $y = f(x)$  in the  $xy$ -plane is the point where the graph intersects the  $x$ -axis, or the point on the graph where the value of  $f(x)$  is equal to 0. Substituting 0 for  $f(x)$  in the equation  $f(x) = 4x + 19$  yields  $0 = 4x + 19$ . Subtracting 19 from each side of this equation yields  $-19 = 4x$ . Dividing each side of this equation by 4 yields  $x = -\frac{19}{4}$ . Therefore, when  $f(x) = 0$ , the number is  $-\frac{19}{4}$ .

Choice B is incorrect. This is the best interpretation of the  $y$ -intercept, not the  $x$ -intercept, of the graph of the function.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the best interpretation of the slope, not the  $x$ -intercept, of the graph of the function.

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## Q4

16.25

Accept also:  $65/4$

The correct answer is  $\frac{65}{4}$ . The linear relationship between  $x$  and  $y$  can be represented by the equation  $y = mx + b$ , where  $m$  and  $b$  are constants. It's given in the table that when  $x = -12$ ,  $y = -45$ . Substituting  $-12$  for  $x$  and  $-45$  for  $y$  in the equation  $y = mx + b$  yields  $-45 = -12m + b$ , which can be rewritten as  $-45 + 12m = b$ . It's also given in the table that when  $x = 6$ ,  $y = 45$ . Substituting  $6$  for  $x$  and  $45$  for  $y$  in the equation  $y = mx + b$  yields  $45 = 6m + b$ , which can be rewritten as  $45 - 6m = b$ . Substituting  $-45 + 12m$  for  $b$  in this equation yields  $45 - 6m = -45 + 12m$ . Adding  $6m$  to both sides of this equation yields  $45 = -45 + 18m$ . Adding  $45$  to both sides of this equation yields  $90 = 18m$ . Dividing both sides of this equation by  $18$  yields  $5 = m$ , or  $m = 5$ . Substituting  $5$  for  $m$ ,  $-12$  for  $x$ , and  $-45$  for  $y$  in the equation  $y = mx + b$  yields  $-45 = 5(-12) + b$ , or  $-45 = -60 + b$ . Adding  $60$  to both sides of this equation yields  $15 = b$ . Therefore,  $m = 5$  and  $b = 15$ . Substituting  $5$  for  $m$  and  $15$  for  $b$  in the equation  $y = mx + b$  yields  $y = 5x + 15$ . Thus, the equation  $y = 5x + 15$  represents the linear relationship between  $x$  and  $y$ . It's also given that the graph of the linear equation representing this relationship passes through the point  $\frac{1}{4}, a$ . Substituting  $\frac{1}{4}$  for  $x$  and  $a$  for  $y$  in the equation  $y = 5x + 15$  yields  $a = 5\frac{1}{4} + 15$ , which is equivalent to  $a = \frac{5}{4} + 15$ , or  $a = \frac{65}{4}$ . Note that  $65/4$  and  $16.25$  are examples of ways to enter a correct answer.

## Q5

D

D. Kaylani purchased 6 yards more fabric than she used to make the suits.

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Choice D is correct. It's given that the equation  $y - 5x = 6$  represents the relationship between the number of suits that Kaylani made,  $x$ , and the total length of fabric she purchased,  $y$ , in yards. Adding  $5x$  to both sides of the given equation yields  $y = 5x + 6$ . Since Kaylani made  $x$  suits and used 5 yards of fabric to make each suit, the expression  $5x$  represents the total amount of fabric she used to make the suits. Since  $y$  represents the total length of fabric Kaylani purchased, in yards, it follows from the equation  $y = 5x + 6$  that Kaylani purchased  $5x$  yards of fabric to make the suits, plus an additional 6 yards of fabric. Therefore, the best interpretation of 6 in this context is that Kaylani purchased 6 yards more fabric than she used to make the suits.

Choice A is incorrect. Kaylani made a total of  $x$  suits, not 6 suits.

Choice B is incorrect. Kaylani purchased a total of  $y$  yards of fabric, not a total of 6 yards of fabric.

Choice C is incorrect. Kaylani used a total of  $5x$  yards of fabric to make the suits, not a total of 6 yards of fabric.

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**Q6****B**

B.  $hx = 5x + 7$

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Choice B is correct. An equation that defines a linear function  $f$  can be written in the form  $fx = mx + b$ , where  $m$  and  $b$  are constants. It's given in the table that when  $x = -4$ ,  $fx = 0$ . Substituting  $-4$  for  $x$  and  $0$  for  $fx$  in the equation  $fx = mx + b$  yields  $0 = m(-4) + b$ , or  $0 = -4m + b$ . Adding  $4m$  to both sides of this equation yields  $4m = b$ . Substituting  $4m$  for  $b$  in the equation  $fx = mx + b$  yields  $fx = mx + 4m$ . It's also given in the table that when  $x = -\frac{19}{5}$ ,  $fx = 1$ . Substituting  $-\frac{19}{5}$  for  $x$  and  $1$  for  $fx$  in the equation  $fx = mx + 4m$  yields  $1 = m(-\frac{19}{5}) + 4m$ , or  $1 = -\frac{19}{5}m + 4m$ . Multiplying both sides of this equation by  $5$  yields  $5 = -19m + 20m$ , or  $5 = m$ . Substituting  $5$  for  $m$  in the equation  $fx = mx + 4m$  yields  $fx = 5x + 20$ . If  $hx = fx - 13$ , substituting  $5x + 20$  for  $fx$  in this equation yields  $hx = 5x + 20 - 13$ , or  $hx = 5x + 7$ .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is an equation that defines the linear function  $f$ , not  $h$ .

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#### Notes

Accept equivalent forms (e.g.  $0.25 \equiv 1/4$ ). For grid-in items, decimal and fraction answers are both valid.

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